The science behind the swim

This animation is driven by measured biomechanics, not by keyframing. This document records every piece of research it rests on, what each parameter is set to and why, and — as importantly — where the published evidence runs out and a choice had to be made instead.

Three things are worth stating at the top, because they shaped everything below:

1. **The obvious model was wrong three times.** The head moves *with* the flukes, not against them; the amplitude minimum sits at the flippers, not at the nose; and a cetacean blink is a corner-anchored purse, not an upper-lid shutter. Each is counter-intuitive, each is measured, and each is built in.
2. **The detailed kinematic dataset is for dolphins, not humpbacks.** The per-body-point amplitude and phase numbers that make the whole animal move correctly were measured on odontocetes (dolphins, orca, beluga). For humpbacks there is exactly *one* direct fluke-amplitude measurement in the literature. Everything humpback-specific below is frequency, speed and body shape. Where an odontocete number is used for a humpback, it is flagged.
3. **Nothing here is invented to look scientific.** Two numbers in the whole build are unsourced — the blink interval and duration — because no measurement of them exists for any cetacean. They are labelled as artistic choices wherever they appear.

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1. Body length, speed and tail-beat frequency

These three set the fundamental scale and timing of the animation. All are from tag data on live baleen whales.

Body length — 13.5 m.

> *Woodward, B. L., Winn, J. P. & Fish, F. E. (2006). "Morphological specializations of baleen whales associated with hydrodynamic performance and ecological niche."* Journal of Morphology *267: 1284–1294. doi:10.1002/jmor.10474*

Adult humpback mean length 13.50 m (n = 128, whaling and stranding records; range 11.58–15.85 m). The same paper places the humpback at fineness ratio 4.21 against the blue whale's 6.37 — i.e. a fast, manoeuvrable body, not a slender stiff cruiser. That matters for §3.

Cruising speed — 2.0 m/s.

> *Gough, W. T., Segre, P. S., Bierlich, K. C., Cade, D. E., Potvin, J., Fish, F. E., et al. & Goldbogen, J. A. (2019). "Scaling of swimming performance in baleen whales."* Journal of Experimental Biology *222: jeb204172. doi:10.1242/jeb.204172*

Humpback tag data (n = 97 deployments): mean speed 1.77 m/s; a photogrammetry subset gives 1.99 ± 0.45 m/s, with a modelled optimal speed of 1.95 ± 0.04 m/s. A companion paper (Gough et al. 2021, *JEB* 224: jeb237586) gives routine swimming at 2.09 ± 0.066 m/s. **2.0 m/s sits in the centre of all of these.**

Locally relevant cross-check, migrating humpbacks off the Gold Coast:

> *Kela, H., de Bie, J., Paas, K. H. W., Stack, S., Franklin, W., Franklin, T. & Meynecke, J.-O. (2024). "Assessment of humpback whale swimming speeds in two eastern Australian bays."* Marine and Freshwater Research. *doi:10.1071/MF24116*

Mean 1.15 m/s in the Gold Coast bay (a minimum-speed method, so it under-reads). The animation's 2.0 m/s is the open-water cruising figure, not the in-bay one.

Tail-beat frequency — ~0.23 Hz, i.e. one beat every ~4.3 seconds. From Gough et al. 2019: humpback mean 0.229 ± 0.039 Hz (n = 97). This is *the* number that separates a whale from a fish on screen — almost every hand-animated whale beats its tail roughly ten times too fast. In the animation the frequency is not set directly; it falls out of the Strouhal relation (§2).

Frequency scales with body length as $L^{-0.53}$ (Gough et al. 2019, log-log regression, $R^2 = 0.635$), which is far shallower than the $L^{-1.0}$ an optimal-Strouhal or resonant-beam oscillator predicts. The authors' explicit conclusion is that whales are **not** operating as simple Strouhal machines. This is why the Strouhal lock below is described as an animation control rather than a law the animal obeys.

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2. Strouhal number and fluke amplitude

Fluke-beat amplitude — 0.20 of body length, peak-to-peak.

The field-standard assumption is A = L/5:

> *Bainbridge, R. (1958). "The speed of swimming of fish as related to size and to the frequency and amplitude of the tail beat."* Journal of Experimental Biology *35: 109–133. ("This maximum amplitude is the same for all fish tested and is about one-fifth of the body length.")*

For cetaceans specifically it is validated by the one direct humpback measurement, from a tag that slipped onto a fluke blade (Gough et al. 2019): peak-to-peak amplitude **2.63 ± 0.79 m**, which on an ~11–13 m animal is A/L ≈ 0.20–0.24. Odontocete measurements agree:

> *Rohr, J. J. & Fish, F. E. (2004). "Strouhal numbers and optimization of swimming by odontocete cetaceans."* Journal of Experimental Biology *207: 1633–1642. (n = 248 measurements, 6 species; peak-to-peak fluke amplitude / body length "occurred predominantly between 0.15 and 0.25".)*

Amplitude is **independent of swimming speed** across all these studies — speed is modulated by *frequency*, not by how hard the tail swings. The animation respects this: the amplitude parameter is fixed and speed changes drive frequency.

The Strouhal relation — St = fA/U. The animation's default St = 0.31 is chosen so the lock reproduces the *measured frequency* (0.23 Hz) at the measured speed and amplitude. The efficient band is usually quoted as 0.25–0.35, but the cetacean data is broader and lower:

- The one directly measured humpback Strouhal was **0.24 ± 0.04** (Gough et al. 2019), at a faster 2.61 m/s.
- Rohr & Fish 2004 (n = 248): "most cetaceans prefer swimming at St values between 0.20 and 0.40"; 74% fall in 0.20–0.30; only 55% fall inside the theoretical 0.25–0.35 band.

The animation's on-screen readout warns below St 0.20 or above 0.35, using the wider cetacean-measured range rather than the textbook one.

An honest tension. A/L = 0.20 *and* U = 2.0 m/s *and* f = 0.23 Hz together imply St ≈ 0.31, which is inside the efficient band but above the single measured humpback value of 0.24 (which came at 2.61 m/s). These cannot all be honoured at one speed. The frequency is the better-evidenced number (n = 97 deployments versus n = 1 animal for St), so the lock is tuned to reproduce it.

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3. The whole animal moves — the amplitude envelope and phase

This is the core of the realism, and where the intuitive model fails hardest.

The load-bearing source

> *Fish, F. E., Peacock, J. E. & Rohr, J. J. (2003). "Stabilization mechanism in swimming odontocete cetaceans by phased movements." Marine Mammal Science 19(3): 515–528.*

They digitised four points — rostrum, flipper tip, peduncle, fluke tip — across seven species at 1.4–7.3 m/s. The results, and how each is built in:

Point	Measured amplitude	Measured phase vs fluke	In the animation
Rostrum (nose)	0.02–0.06 L (fluke is 4.34–6.76× larger)	**−9.4° to +33.0°**	19% of fluke amplitude, +20°
Flipper tip	**smallest of the four**	trails by 60.9–123.4°	smallest excursion; trails 90°
Peduncle	between	**leads** by 18.9–48.7°	leads ~38°
Fluke tip	0.17–0.25 L	(reference)	reference

Verified output of the animation, measured off the live page:

	model	measured target
Rostrum amplitude	19% of fluke, 3.8% L	15–23%, 2–6% L
Rostrum phase	+20°	−9.4° to +33.0°
Peduncle phase	leads 38°	leads 18.9–48.7°
Flipper excursion	smallest of four	smallest of four

The two counter-intuitive findings

(a) The head moves nearly IN phase with the flukes, not against them. A rigid beam pivoting about its centre of mass would put rostrum and fluke 180° apart. Real animals do not: in two of the seven species the phase difference was statistically indistinguishable from zero. Fish et al. verbatim:

> *"By synchronously moving the rostrum in the same direction as flukes by muscular control, there is a reduction in the natural tendency of the rostrum to swing through a wide arc opposite the motion of the flukes."*

So the animation models the body in **two zones meeting at a node**: a travelling wave behind the node, and an anterior section whose phase is *prescribed against the fluke* rather than inherited from the wave. A single travelling wave cannot produce a near-synchronous rostrum and a peduncle that leads by only ~38° at the same time; the split is what the measured muscular stabilisation actually does.

(b) The amplitude minimum is at the flippers, not the nose. The flipper tip has the smallest excursion of the four points, because it sits nearest the centre of mass. So the envelope is not a monotonic ramp from a still nose to a big tail — it dips in the middle. In this sprite the measured node lands between the two flipper hinges ($u = 0.25$ and $u = 0.37$), an independent check on the earlier segmentation.

Independent corroboration across 44 species

> *Di Santo, V., Goerig, E., Wainwright, D. K., Akanyeti, O., Liao, J. C., Castro-Santos, T. & Lauder, G. V. (2021). "Convergence of undulatory swimming kinematics across a diversity of fishes." PNAS 118(49): e2113206118.*

Global fit across 44 species (peak-to-peak, proportion of body length): $\mathbf{y = 0.05 - 0.13x + 0.28x^2}$ — head 0.05 L, tail 0.20 L, so head:tail = 25%, with a minimum at $x = 0.232$ L. Their across-species medians give head:tail ≈ 17%. The animation sits at **19%** with the node at 0.28 — between their fit and their medians, arrived at from an entirely separate (cetacean) literature. Two independent lines converging is the strongest single piece of evidence here.

Di Santo et al. also demolish the textbook ordering: "there was **no decrease in head:tail amplitude from the anguilliform to thunniform mode** of locomotion as we expected from the traditional classification," and the ratio "was only statistically higher in the tuna." The supposedly stiffest swimmer has the *most* head yaw. A locked head is not conservative realism — it is a myth.

The same quadratic envelope traces back to fish kinematics measured by:

> *Videler, J. J. & Hess, H. (1984), on steadily swimming saithe and mackerel; as fitted in Borazjani, I. & Sotiropoulos, F. (2010), "On the role of form and kinematics on the hydrodynamics of self-propelled body/caudal fin swimming," Journal of Experimental Biology 213: 89–107 — envelope $a(z) = 0.02 - 0.08z + 0.16z^2$, giving nose = 20% of tail and a node at $z = 0.25$. A third independent arrival at the same ~20% / node-at-0.25 answer.*

Lighthill recoil — implemented, but off by default

> *Lighthill, M. J. (1960). "Note on the swimming of slender fish." Journal of Fluid Mechanics 9(2): 305–317. Lighthill, M. J. (1970). "Aquatic animal propulsion of high hydromechanical efficiency." Journal of Fluid Mechanics 44(2): 265–301.*

A prescribed body wave carries net lateral momentum and net angular momentum, which a free-swimming body cannot. Requiring both to be zero leaves a rigid-body heave and pitch — the "recoil" —

that get subtracted from the wave. Lighthill 1970 (abstract): the recoil movements are "minimized with the right distribution of total inertia (the sum of fish mass and the water's virtual mass)," which is why the animation weights the momentum balance by the measured thickness profile squared (a slender body's mass and added mass both scale with cross-section).

The recoil solver is implemented and, when enabled, roughly doubles the head's throw and swings it into anti-phase. **It defaults to zero**, for two reasons:

1. The measured phases (§3a) say a live cetacean muscularly *suppresses* that anti-phase swing.
2. More decisively, measured envelopes already contain whatever recoil was not cancelled, so adding a recoil term on top double-counts it. This is made explicit in the modelling literature:

   > *Maertens, A. P., Gao, A. & Triantafyllou, M. S. (2017). "Optimal undulatory swimming for a single fish-like body and for a pair of interacting swimmers." Journal of Fluid Mechanics 813: 301–345.*

who set their recoil term B(x,t) = 0 for the canonical carangiform gait *precisely because* the envelope came from Videler's measurements.

The slider is kept so the anti-phase recoil the animal works to suppress can be seen; it is not meant to be used for the finished look.

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4. The pectoral flippers

> *Fish, F. E. & Battle, J. M. (1995). "Hydrodynamic design of the humpback whale flipper." Journal of Morphology 225(1): 51–60.*

Humpback flippers are exceptionally long (0.28–0.33 of body length), aspect ratio ~6, with 19° sweepback and tubercled leading edges. In this build they are cut as separate layers with hinges measured at their roots — the near flipper composited in front of the body, the far flipper behind.

What flippers do during steady swimming — the governing paradigm:

> *Segre, P. S., Seakamela, S. M., Meÿer, M. A., Findlay, K. P. & Goldbogen, J. A. (2017). "A hydrodynamically active flipper-stroke in humpback whales." Current Biology 27(13): R636–R637.*
>
> *"A central paradigm of aquatic locomotion is that cetaceans use fluke strokes to power their swimming while relying on lift and torque generated by the flippers to perform maneuvers such as rolls, pitch changes and turns."*

Segre et al. document an *active* flapping flipper-stroke — but only immediately before mouth-opening during lunge feeding, observed twice in hundreds of hours of video. In cruising, the flippers are control surfaces, not propulsors. The only descriptive study of live humpback flipper motion agrees:

> *Edel, R. K. & Winn, H. E. (1978). "Observations on underwater locomotion and flipper movement of the humpback whale Megaptera novaeangliae." Marine Biology 48: 279–287.*

which reports large flipper movements reserved for banked turns above ~4 knots, not for straight cruising.

How the flippers move here. Fish et al. 2003 (§3) measured the flipper tip trailing the flukes by 60.9–123.4° with the smallest excursion of the four points. The animation drives them off that measured phase lag (default 90°) with deliberately small amplitude, opposing pitch. This is the honest position, because:

> ***No one has published a flipper motion time series for steady straight-line cruising, for any cetacean.*** *Sweep, feathering angle and dihedral over a cruise cycle are simply not in the literature.*

So the flipper motion is a physically-motivated extrapolation from the one measured quantity (phase lag), not a reproduction of measured kinematics. It is kept subtle for exactly that reason.

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5. Burst-and-coast, and duty-compensated thrust

Large whales rarely beat continuously; they stroke a few times and then glide while drag bleeds the speed off. This is standard cetacean swimming behaviour (described throughout Fish & Rohr's work). The animation defaults to 2 beats then a 3-second glide — about two full cycles per 24-second crossing, beating 41% of the time.

The one subtlety worth recording: thrust is scaled by **1/duty** so the burst/glide sliders change the *rhythm* without changing the *speed*. Calibrating thrust for continuous beating (thrust = drag·U^2) undershoots at any duty below 1; because frequency is Strouhal-locked to instantaneous speed, that shortfall would otherwise drag the beat rate out of the measured band as a side effect of a purely rhythmic control. Mean thrust must equal mean drag, so the instantaneous value during a burst is drag·U^2/duty. Verified on the live page: speed oscillates 1.72–2.28 m/s and the beat 0.198–0.262 Hz across every rhythm setting, every sample inside the measured 0.229 ± 0.039 Hz band.

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6. The blink

> *Nishimaniwa, K., Yamada, T. K., Sekiya, S. I., Amano, M. & Tajima, Y. (2026). "Morphological features of the orbicularis oculi muscle and facial nerve in four odontocete families, with comparisons within Cetartiodactyla." Journal of Veterinary Medical Science 88(1): 1–12.*

A cetacean blink is **a purse, not a shutter**. The orbicularis oculi closes the palpebral fissure "using the medial and lateral canthi as fulcrums" — a sphincter contraction anchored at the corners, with the retractor bulbi pulling the globe inward at the same time. In the animation both margins converge on a mid-line, the travel tapers to zero at the corners, and the closing line bows.

Humpbacks are the awkward case, and the animation handles it deliberately. In dolphins and porpoises the upper palpebral region has degenerated into an aponeurotic sheet with almost no facial-nerve supply, so the lower lid does most of the moving. Mysticetes did not degenerate:

> *Rodrigues, F. M. et al. (2015). "Morphology of accessory structures of the humpback whale eye (Megaptera novaeangliae)." Acta Zoologica 96: 328–334. Zhu, Q., Hillmann, D. J. & Henk, W. G. (2000). On the bowhead whale eye, Anatomical Record 259: 189–204.*

In humpback and bowhead the muscle is "entirely composed of muscular fibers," the ancestral mammalian arrangement — so a humpback blink is *more* symmetric than a dolphin's. Both lids therefore move by equal measure here.

No nictitating membrane. Cetaceans lack one:

> *Meshida, K., Lin, S., Domning, D. P., Reidenberg, J. S., Wang, P. & Gilland, E. (2020). "Cetacean Orbital Muscles: Anatomy and Function of the Circular Layers." Anatomical Record.*

which lists the nictitating membrane among features "absent in cetaceans but present in other closely related terrestrial mammals." Most popular sources claim whales have a third eyelid; they are wrong, and none is animated.

Closure is faster than reopening — this asymmetry *is* sourced:

> *Aiello, B. R. et al. (2023). "The origin of blinking in both mudskippers and tetrapods is linked to life on land." PNAS 120(18): e2220404120.*

Peak eye velocity and acceleration significantly higher during closure than reopening ($P < 10^{-19}$), and the asymmetry "is observed in all tetrapods for which data is available." The animation's blink stroke is fast-down, brief-hold, slower-up accordingly.

The two unsourced numbers

There is no published measurement of blink rate or blink duration for any cetacean species. This was checked systematically against Europe PMC across species and term variants; every cetacean hit is either anatomy with no kinematics, or sleep work that scores eye state as a sustained condition in 5-second epochs rather than counting blinks (e.g. Lyamin, Mukhametov & Siegel 2004, *Archives Italiennes de Biologie* 142: 557–568).

So the mean interval (11 s) and stroke duration (0.55 s) are **artistic choices** — the only two numbers in the entire build not backed by a measurement. The literature licenses "infrequent and slow": the 5-second sleep-scoring methodology only works if frequent blinks are not happening, and cetaceans have well-developed Harderian glands, so blinking "seems unnecessary for maintaining the moisture of the corneal surface" (Nishimaniwa et al.). That is an argument for a slow, rare blink, not a measurement of one.

Widely-repeated web claims that whales "blink once every couple of hours" have no citation and no traceable origin. They are not used.

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7. Body flexibility — why this is modelled as more than a stiff tail

The choice to move the whole animal, rather than confining motion to the rear third, is grounded in the humpback being genuinely more flexible than the dolphins the §3 kinematics come from.

> *Buchholtz, E. A. (2001). "Vertebral osteology and swimming style in living and fossil whales (Order: Cetacea)." Journal of Zoology 253(2): 175–190.*

Mysticetes retain the archaeocete pattern of near-constant vertebral shape along the torso, which acts as a single undulatory unit. As quoted in Kot et al. (2022, *Integrative Organismal Biology* 4(1): obab036), dorsoventral displacement "began at the chest in a swimming humpback whale," versus being "largely restricted to the caudal peduncle" in porpoises and white-sided dolphins.

The consequence for this animation is important and honest: because humpbacks are *more* flexible than the odontocetes measured in §3, the anterior share of the motion here is plausibly **under**-stated, not over-stated. There is no published humpback per-body-point number to correct it with, so the odontocete value is used as-is rather than invented upward.

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8. Summary of parameters and their provenance

Parameter	Value	Source	Confidence
Body length	13.5 m	Woodward et al. 2006 (n=128)	Humpback, measured
Cruising speed	2.0 m/s	Gough et al. 2019/2021	Humpback, measured
Tail-beat frequency	~0.23 Hz	Gough et al. 2019 (n=97)	Humpback, measured
Fluke amplitude	0.20 L p-p	Gough et al. 2019; Bainbridge 1958	Humpback (n=1) + convention
Strouhal number	0.31 (tuned to f)	Rohr & Fish 2004; Gough 2019	Reproduces measured f
Rostrum amplitude	19% of fluke	Fish, Peacock & Rohr 2003	Odontocete, measured
Rostrum phase	+20° vs fluke	Fish, Peacock & Rohr 2003	Odontocete, measured
Peduncle phase	leads ~38°	Fish, Peacock & Rohr 2003	Odontocete, measured
Node position	u = 0.28	Fish 2003; Di Santo 2021; Videler 1984	Three independent lines
Flipper phase lag	90° trailing	Fish, Peacock & Rohr 2003	Odontocete, measured
Flipper amplitude	small	Segre 2017; Edel & Winn 1978	Descriptive only
Blink mechanism	corner-anchored purse, both lids	Nishimaniwa 2026; Rodrigues 2015	Anatomy, measured
Blink close vs open	faster close	Aiello et al. 2023	Measured (all tetrapods)
Blink interval	11 s	—	**Artistic choice — no data exists**
Blink duration	0.55 s	—	**Artistic choice — no data exists**

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9. Where the animation is honestly limited

- **Most per-body-point kinematics are odontocete, not humpback.** The head and peduncle phase relations were measured on dolphins, orca and beluga. They are the best available and are almost certainly conservative for a humpback (§7), but they are not humpback measurements.
- **Flipper cruise motion is extrapolated, not reproduced.** No time series exists for any cetacean; only the phase lag is measured (§4).
- **The blink timing is unsourced** because no cetacean measurement exists (§6).
- **A flat sprite cannot rotate in depth.** The source is a three-quarter view, so the fluke's angle of attack is faked by the phase lag rather than being a real pitch. It holds at the default amplitude and reads as rubber if pushed far past it.
- **The Strouhal lock is a control, not a law.** Real whales scale frequency as $L^{-0.53}$, not the $L^{-1.0}$ the lock implies (§1). It is kept because it retimes the tail correctly when speed changes, not because the animal obeys it.

Full reference list

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Compiled from three literature reviews conducted during the build. Every value above was traced to the cited work; where a source could only be reached through a secondary citation (e.g. Videler & Hess 1984, whose fits are quoted via Borazjani & Sotiropoulos 2010), that is stated inline. The distinction between measured humpback data, measured odontocete data used for a humpback, and unsourced artistic choices is maintained throughout — it is the difference between "true to life" and "looks about right," and this animation tries to be honest about which is which.